



Genium INET[®]

OUCH Protocol Specification

OSE



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1 Summary Of Changes

Changes between this version and version a2614 (4.1.1430).

No	Changes	Comment
1	Enter MM Order on page 9 added.	
2	Mass Cancel on page 13 added.	
3	Mass Cancel Accepted on page 14 added.	
4	Order Replaced on page 15 : Field description for Price improved.	
5	Order Executed on page 17 : - Field description for Match ID and Combo Group ID improved. - Length of field Deal Source corrected to 1.	
6	Note about implicit decimals for Numeric Unsigned Data Type removed in Data Types on page 7 .	
7	Term "Suspended" replaced by term "Inactivated" in Order Canceled on page 16 .	
8	New section Login Limitation on page 6 added.	

2 About the Manual

The purpose of this document is to describe the OUCH protocol.

2.1 References

For more information, refer to the following documents:

- *Genium INET ITCH Protocol Specification*
- *System Error Message Reference*

3 OUCH Overview

The OUCH protocol accepts limit orders from system participants and executes matching orders when possible. Non-matching orders may be added to the order book where they are waiting to be matched according to the matching priority model.

OUCH is a simple protocol that allows Genium INET Trading users to enter orders, replace and cancel existing orders and receive executions. It is intended to allow participants and their software developers to integrate Genium INET Trading into their proprietary trading systems or to build custom front ends.

OUCH only provides a method for participants to send orders to Genium INET Trading and receive updates on those orders entered. For information about all orders entered into and executed on the Genium INET Trading book, refer to the ITCH protocol specification.

OUCH is the low-level native protocol for connecting to the Genium INET Trading system. It is designed to offer the maximum possible performance at the cost of flexibility and ease of use. For applications that do not require this extreme level of performance, Genium INET Trading offers other, more standard interfaces that may be more suitable and easier to develop to.

3.1 Architecture

The OUCH protocol is composed of logical messages passed between the OUCH host and the client application. Each message type has a fixed message length. The messages are binary encoded, which means that all numeric values are represented as binary values. Character or string values are composed of non-control ISO 8859-1 (Latin-1) encoded bytes.

All (outbound) messages sent from the OUCH system to the client are assumed to be sequenced, and their delivery is guaranteed by the lower level protocol. The SoupBinTCP protocol (specification available separately) is used to guarantee the delivery and sequencing of OUCH messages sent from the host to the client. Please refer to the SoupBinTCP protocol specification for details.

Messages sent from the OUCH client to the host are inherently non-guaranteed, even if they are carried by a lower level protocol that guarantees delivery (like TCP/IP sockets). Therefore, all host-bound messages are designed so that they can be benignly resent for robust recovery from connection and application failures.

Each physical OUCH host port is bound to an OUCH Account assigned by the marketplace. On a given day, every order entered on OUCH is uniquely identified by the combination of the logical OUCH Account and the participant-created Token field.

3.1.1 Login Limitation

Since $2^{63} - 1$ is the maximum number that can be represented with 64-bit two's complement integer, which is the largest primitive integer that can be represented in java, the maximum sequence number the OUCH Gateway (DE_GWY) can interpret is $2^{63} - 1$.

This limitation is in the OUCH Gateway and not in the SoupBinTCP protocol.

3.2 Data Types

All Integer fields are composed of binary encoded numbers.

Table 1:

Type	Size	Notes
Numeric unsigned	1, 2, 4, 8 or 12 bytes	Unsigned big-endian binary encoded numbers.
Numeric signed	1 or 4 bytes	Signed big-endian binary encoded numbers.
Alpha	variable	Left justified and padded on the right with spaces. Composed of non-control ISO 8859-1 (Latin-1) encoded bytes.
Price	4 bytes	Prices are signed integer fields. Number of decimals and allowed tick steps are specified in the ITCH Order book Directory message (Number of decimals in Price).
Timestamp	8 bytes	UNIX Time (number of nanoseconds since 1970-01-01 00:00:00 UTC)

3.3 Fault Redundancy

A single OUCH Account can be bound to two physical OUCH gateways. These OUCH gateways are configured as one primary and one standby for fault redundancy. In this configuration the standby node will open for connections automatically, if the primary node fails.

4 Message Formats

4.1 Inbound Messages

Inbound messages are sent from the participant's application to the OUCH host. They are not sequenced. All Inbound Messages may be repeated benignly. This gives the client the ability to resend any inbound message if, in the case of a connection loss or an application error, it is uncertain whether or not the Genium INET Trading system received it.

The idea of benign inbound message retransmission with end-to-end acknowledgement is fundamental to OUCH's fail-over redundancy. If your connection ever fails, there is no way for you to know if pending messages actually made it over the link before the failure. A robust OUCH client can safely resend any pending messages over a mirrored link without worrying about generating duplicates. This applies to Genium INET Trading's disaster failover capability as well; if the system ever needs to fail over to the backup site, some messages sent at the moment of the failure may be lost. A robust application can simply resend the pending messages, making the failover seamless to the end user.

4.1.1 Enter Order

Enter Order is used to enter a new order into the system. The response to a successful Enter Order is an Order Accepted message. If the order is rejected, the Order Rejected message will be returned.

Enter Order can be updated with the Replace Order message.

A client must submit a client-generated order id called the Order Token on each new order. This token is returned on subsequent order acks and order updates.

Single series and Combination series can be entered with the Enter Order message.

Note

All orders entered via OUCH will default to the Round Lot type. The quantity of the order must be a multiple of the Round Lot Size. The Round Lot Size of a particular order book can be found in the ITCH Order book Directory message.

Table 2: Enter Order Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"O"	Enter Order Message.
Order Token	1	14	Alpha	Client-generated order identifier.
Order book ID	15	4	Numeric signed	Order book identifier. The Order book ID can be obtained by ITCH.
Side	19	1	Alpha	Values: "B" = Buy order "S" = Sell order
Quantity	20	8	Numeric unsigned	Quantity.
Price	28	4	Price	Signed integer price.

Name	Offset	Length	Value	Notes
Time In Force	32	1	Numeric unsigned	Values: 0 = Day 3 = Immediate or Cancel (FaK) 4 = Fill or Kill
Open Close	33	1	Numeric unsigned	Position update for the account. Values: 0 = Default for the account 1 = Open 2 = Close/Net 3 = Mandatory close
Client/Account	34	16	Alpha	Pass-thru field. Same usage as for ex_client_s of OMnet API.
Customer Info	50	15	Alpha	Pass-thru field.
Exchange Info	65	32	Alpha	The Exchange info field is formatted and validated according to the OMNet overlay structure OSE_EXCHANGE_INFO

4.1.2 Enter MM Order

Enter MM Order is used by market makers (MM) to enter quotes into Market. Similar to the Enter Order message, responses are either the Order Accepted message or the Order Rejected message, and Enter MM Order can be updated with the Replace Order message.

Table 3: Enter Market Maker Order Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"P"	Enter MM Order Message.
Order Token	1	14	Alpha	Client-generated order identifier.
Order book ID	15	4	Numeric signed	Order book identifier. The Order book ID can be obtained by ITCH.
Side	19	1	Alpha	Values: "B" = Buy order "S" = Sell order
Quantity	20	8	Numeric unsigned	Quantity.
Price	28	4	Price	Signed integer price.
Client/Account	32	16	Alpha	Pass-thru field. Same usage as for ex_client_s of OMnet API.
Exchange Info	48	32	Alpha	The Exchange info field is formatted and validated according to the OMNet overlay structure OSE_EXCHANGE_INFO

4.1.3 Replace Order

Replace Order is used to modify an existing order entered via OUCH.

Modification of some order parameters may not be allowed, depending on how the system is configured. An Order Rejected will be returned in such case.

There are two order tokens in the Replace message.

- **Existing Order Token**

The Existing Order Token is used to reference the order to be replaced. The Order Token should be from the original Enter Order, not from any intermediate replaces. The current implementation allows intermediate tokens to be used, but this may not be supported in the future.

- **Replacement Order Token**

The Replacement Order Token is the new Order Token that will be assigned to the order if the replace is successful. The replacement Order Token must not be a token previously used in Enter Order or Replace Order transactions.

The response to a Replace Order is:

- **Order Replaced** if the modification was successful. The Order Replaced will contain the current state of the returned order. See below for a discussion on order quantities.
- **Order Rejected** if the replace failed.

It is not possible for OUCH Replace Order message to replace orders entered with OMN API.

Table 4: Replace Order Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"U"	Replace Order Message.
Existing Order Token	1	14	Alpha	The Existing Order Token is used to reference the order to be replaced. The Order Token should be the original Order Token, or the latest successful Replacement Order Token, not any intermediate Replacement Order Token. <i>Note</i> The current implementation allows intermediate tokens to be used, but this may not be supported in the future.
Replacement Order Token	15	14	Alpha	
Quantity	29	8	Numeric unsigned	Desired Open Quantity of the order, in relation to the original order quantity.
Price	37	4	Price	Setting Price to INT_MIN means "no change." Participant can update combo price to Opt via Replace Order.
Open Close	41	1	Numeric unsigned	Position update for the account. Values: 0 = No change 1 = Open 2 = Close/Net 3 = Mandatory close 4 = Default for the account
Client/Account	42	16	Alpha	Pass-thru field. Same usage as for ex_client_s of OMnet API.

Name	Offset	Length	Value	Notes
Customer Info	58	15	Alpha	Pass-thru field.
Exchange Info	73	32	Alpha	The Exchange info field is formatted and validated according to the OMNet overlay structure OSE_EXCHANGE_INFO

4.1.3.1 Order Quantities

In Genium INET OUCH Replace messages, the Quantity field contains the desired total quantity of the order (Order Quantity = open quantity + executed quantity, where open quantity is the quantity of the order in the order book).

Example 1:

1. An order with a quantity of 1000 is entered via OUCH. An Order Accepted with Quantity = 1000 will be returned.
2. A partial execution for 200 occurs. A quantity of 800 is left in the order book. An Executed Order with Traded Quantity = 200 will be returned.
3. The client wants to decrease the open quantity (quantity in the book) to 750. He sends in an Order Replace with Quantity = 950. A Replaced Order with Quantity = 750 will be returned.

Example 2:

1. An order with a quantity of 1000 is entered via OUCH. An Order Accepted with Quantity = 1000 will be returned.
2. The client wants to decrease the open quantity with 600, that is, from the current 1000 to 400. He sends in an Order Replace with Quantity = 400, as a quantity update always is always done in relation to the original order quantity.
3. Simultaneously, 500 of the order is (partially) traded, the trade occurred before the Order Replace. An Executed Order with Traded Quantity = 500 will be returned.
4. The client receives an Order Replaced with Quantity = 0 and Order Status = Not on book, since there is no quantity left in the order book.

Example 3:

1. An order with a quantity of 1000 is entered via OUCH. An Order Accepted with Quantity = 1000 will be returned.
2. A partial execution for 200 occurs. A quantity of 800 is left in the order book. An Executed Order with Traded Quantity = 200 will be returned.
3. The client wants to increase the open quantity (quantity in the book) to 1000. He sends in an Order Replace with Quantity = 1200. A Replaced Order with Quantity = 1000 will be returned

4.1.3.2 Retaining Order book Priority

Most order attributes are allowed to change in Genium INET. A replacement of a certain attribute may cause the order to lose priority, depending on what values are modified, how the system is configured, and so on. A detailed description of the circumstances when a replace causes priority loss is out of scope for this document.

In order to increase the likelihood of keeping an order from losing priority as it is changed, all fields that are intended to be left unchanged should be left unset.

- For Numeric Order parameters, this means setting them to 0 (zero).
- For String (Alpha) fields, the first byte should be set to binary zero '\0' to retain previous value.

Note

The system may be configured to reject replaces that would cause a loss of priority.

4.1.4 Cancel Order

Cancel Order can only be used to cancel orders entered via the same OUCH session. The response to a successful Cancel Order is an Order Canceled message. Failed Cancel Order messages are rejected with the Order Reject message.

Note

The Order Token should be from the original Enter Order, not from any intermediate replaces. The current implementation allows intermediate tokens to be used, but this may not be supported in the future.

Inactive orders can not be deleted with Cancel Order Message.

Partial cancels are not supported with the Cancel Order message. Use Order Replace to modify an existing order.

Table 5: Cancel Order Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"X"	Cancel Order Message.
Order Token	1	14	Alpha	-

4.1.5 Cancel by Order ID

Using the system-generated Order ID, this message can be used to cancel any order in the book regardless of over during which session it was inserted.

The response to a successful Cancel by Order ID is an Order Canceled message. Failed Cancel by Order ID messages are rejected with the Order Reject message.

Inactive orders can not be deleted with Cancel by Order ID Message.

Table 6: Cancel By Order ID Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"Y"	Cancel By Order ID Message.
Order book ID	1	4	Numeric signed	
Side	5	1	Alpha	Values: "B" = Buy order "S" = Sell order

Name	Offset	Length	Value	Notes
Order ID	6	8	Numeric unsigned	The identifier assigned to the order by the system.

4.1.6 Mass Cancel

The Mass Cancel message can be used to cancel one or many orders, and can cancel any order in the book under the same participant regardless of over during which session it was inserted.

The response to a successful Mass Cancel message is a single Mass Canceled Accepted message. Failed Mass Cancel messages are rejected with the Order Reject message.

Table 7: Mass Cancel Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"M"	Mass Cancel Message.
Order Token	1	14	Alpha	Client-generated message identifier.
Underlying ID	15	4	Numeric Unsigned	If orders for the underlying instrument are requested to be cancelled, the Underlying ID must be given.
Scope	19	1	Numeric Unsigned	Values: 1 = Participant Only value "1" is allowed. This field cannot be removed.
Client/Account	20	16	Alpha	This field works same way as ex_client_s of OMNet API (MO4).

4.2 Outbound Messages

Outbound messages are generated by the system and sent to the OUCH client.

4.2.1 Order Accepted

This message acknowledges the receipt and acceptance of a valid Enter Order Message. The data fields from the Enter Order Message are echoed back in this message. Note that the accepted values may differ from the entered values for some fields.

Accepted Messages are guaranteed to come before any Executed Messages or Canceled Messages for an order.

However, when the Order State field of an Accepted Message is Not on book (2), no additional messages will be received for that order.

Note

If the order is traded or otherwise changed immediately at entry, the Order Accepted will show the state of the order after any such operations.

Execution messages and/or cancel messages will follow to account for any differences.

Table 8: Order Accepted Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"A"	Order Accepted Message.
Timestamp – Nanoseconds	1	8	Timestamp	UNIX Time (number of nanoseconds since 1970-01-01 00:00:00 UTC).
Order Token	9	14	Alpha	
Order book ID	23	4	Numeric signed	Order book identifier.
Side	27	1	Alpha	Values: "B" = Buy order "S" = Sell order
Order ID	28	8	Numeric unsigned	The identifier assigned to the new order.
Quantity	36	8	Numeric unsigned	Quantity currently open in the book.
Price	44	4	Price	Signed integer price.
Time In Force	48	1	Numeric unsigned	Values: 0 = Day 3 = Immediate or Cancel (FaK) 4 = Fill or Kill
Open Close	49	1	Numeric unsigned	Position update for the account. Values: 0 = No change 1 = Open 2 = Close/Net 3 = Mandatory close
Client/Account	50	16	Alpha	Pass-thru field. Same usage as for ex_client_s of OMnet API.
Order State	66	1	Numeric unsigned	Values: 1 = On book 2 = Not on book
Customer Info	67	15	Alpha	Pass-thru field.
Exchange Info	82	32	Alpha	The Exchange info field is formatted and validated according to the OMNet overlay structure OSE_EXCHANGE_INFO

4.2.2 Mass Cancel Accepted

This message acknowledges the receipt and acceptance of a valid Mass Cancel message.

Table 9: Mass Cancel Accepted Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"M"	Mass Cancel Message.
Timestamp – Nanoseconds	1	8	Timestamp	UNIX Time (number of nanoseconds since 1970-01-01 00:00:00 UTC).
Order Token	9	14	Alpha	

Name	Offset	Length	Value	Notes
Status	23	4	Numeric Unsigned	The status of the mass cancel specifying the number of cancelled orders.

4.2.3 Order Rejected

This message is used to reject Enter Order messages, Cancel Order messages, and Replace Order messages.

Table 10: Order Rejected Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"J"	Order Rejected Message.
Timestamp – Nanoseconds	1	8	Timestamp	UNIX Time (number of nanoseconds since 1970-01-01 00:00:00 UTC).
Order Token	9	14	Alpha	Populated from the incoming order, invalid values will be preserved.
Reject Code	23	4	Numeric signed	Backend Error Code. See <i>System Error Messages Reference</i> for more information.

4.2.4 Order Replaced

This message acknowledges the receipt and acceptance of a valid Replace Order Message. The data fields from the Replace Order Message are echoed back in this message. Note that the accepted values may differ from the entered values for some fields. You will receive one and only one of these two for each replacement.

Like Accepted Messages, Replaced Messages use the Order State field to denote that a replace was accepted and then automatically canceled when the Order State is Not on book.

No further Executed Messages nor Canceled Messages will be received when the replaced order for the Order State is Not on book.

Note

- If an order is modified by the system, the Replacement Order Token will be blanked out (with spaces). The Previous Order Token will contain the current Order Token as entered by the client.
- If the order is traded or otherwise deleted at the time of Order Replace, the Order Replaced message will show the state of the order after any such operations. Execution messages and/or cancel messages will follow to account for any differences.

Table 11: Order Replaced Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"U"	Order Replaced Message.
Timestamp – Nanoseconds	1	8	Timestamp	UNIX Time (number of nanoseconds since 1970-01-01 00:00:00 UTC).

Name	Offset	Length	Value	Notes
Replacement Order Token	9	14	Alpha	
Previous Order Token	23	14	Alpha	
Order book ID	37	4	Numeric signed	Order book identifier.
Side	41	1	Alpha	Values: "B" = Buy order "S" = Sell order
Order ID	42	8	Numeric unsigned	The identifier assigned to the new order.
Quantity	50	8	Numeric unsigned	Quantity currently open in the book.
Price	58	4	Price	Signed integer price.
Time In Force	62	1	Numeric unsigned	Values: 0 = Day 3 = Immediate or Cancel (FaK) 4 = Fill or Kill
Open Close	63	1	Numeric unsigned	Defines the position update for the account. Values: 0 = Default for the account 1 = Open 2 = Close/Net 3 = Mandatory close
Client/Account	64	16	Alpha	Pass-thru field. Same usage as for ex_client_s of OMnet API.
Order State	80	1	Numeric unsigned	Values: 1 = On book 2 = Not on book
Customer Info	81	15	Alpha	Pass-thru field.
Exchange Info	96	32	Alpha	The Exchange info field is formatted and validated according to the OMNet overlay structure OSE_EXCHANGE_INFO

4.2.5 Order Canceled

A Canceled Message informs you that an order has been canceled. This could be acknowledging a Cancel Order Message, or it could be the result of system cancellation of the order.

Note

Order Canceled messages are sent out when orders are inactivated due to connection loss. Orders cannot be re-activated again. However, it is possible to cancel an inactivated order, therefore it is possible to receive more than one Order Canceled message for the same order.

Table 12: Order Canceled Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"C"	Order Canceled Message.
Timestamp – Nanoseconds	1	8	Timestamp	UNIX Time (number of nanoseconds since 1970-01-01 00:00:00 UTC).
Order Token	9	14	Alpha	
Order book ID	23	4	Numeric signed	Order book identifier.
Side	27	1	Alpha	Values: "B" = Buy order "S" = Sell order
Order ID	28	8	Numeric unsigned	The identifier assigned to the canceled order.
Reason	36	1	Numeric unsigned	Contains the system change reason values. Values: Values: 1 = Canceled by user 4 = Order Inactivated 9 = Canceled by system 10 = Canceled by proxy 15 = Canceled by System Limit Change 19 = Deleted by time validity expiration 20 = Canceled due to ISS 21 = Inactivated due to ISS 34 = Canceled after auction 41 = Canceled by System Delta Protection 42 = Canceled by System Quantity Protection 43 = Canceled because trader is not allowed to trade with himself

4.2.6 Order Executed

This message is returned when a partial or full fill occurs. In case of a combination order fill, one Order Executed message will be received per leg.

Table 13: Order Executed Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"E"	Order Executed Message.
Timestamp – Nanoseconds	1	8	Timestamp	UNIX Time (number of nanoseconds since 1970-01-01 00:00:00 UTC).
Order Token	9	14	Alpha	
Order book ID	23	4	Numeric signed	Needed for Combination fills (Order Executed sent per leg).
Traded Quantity	27	8	Numeric unsigned	
Trade Price	35	4	Price	Signed integer trade price. Number of decimals is given by the Order book Directory message in ITCH.

Name	Offset	Length	Value	Notes
Match ID	39	8	Numeric unsigned	Backend-generated identifier. Match ID might not always be unique. When the same Match ID is received the quantities should be aggregated. An ever increasing number per partition, assigned to an execution event. The match id value is equivalent to the OMNet match_id and is set in the same rule as BO5 or BD6.
Combo Group ID	47	4	Numeric unsigned	This field is only populated if the trade is part of combination trading, either a combination order matching another combination order or a combination order matching in the leg (outright) order book. For trades in the single orderbooks, which are not related to combination trading, this field will always be 0 (zero). This number is unsigned.
Deal Source	51	1	Numeric	The deal source is a numeric code equivalent to the OMNet deal_source that gives information about how the execution took place. 1 = Matched by system, automatically. 7 = Combination order matched against another combination order when matched by the Exchange, electronically. 20 = Deal made at the end of an auction.